ABHIJIT ROY

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PROFESSIONAL SUMMARY

B.Tech Information Technology graduate (2026, CGPA 7.69/10) with hands-on experience across the Generative AI model lifecycle — data preparation, RAG framework customization, prompt engineering, tool-calling agent design, and evaluation. Built and deployed multiple end-to-end LLM applications, including CityPilot, a tool-calling AI agent (LangChain, Google Gemini) that autonomously decides between live-weather and news-search tools based on natural-language reasoning; a Retrieval-Augmented Generation (RAG) chatbot that answers questions from user-uploaded PDFs (LangChain, ChromaDB, Mistral AI); and an LLM-powered assistant (Groq API, LLaMA 3.3 70B) grounding responses in live structured data. Comfortable with Python (Pandas, NumPy), SQL, and Streamlit, with a track record of researching model behavior, documenting architectures, and collaborating cross-functionally to ship working AI applications. Seeking an Analytics Analyst role focused on generative AI / LLM model development.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Matplotlib, Streamlit), SQL, Java (Basic), C (Basic), HTML, CSS, JavaScript, React, Node.js, REST API, FastAPI
- **Generative AI & LLMs:** LLM APIs (Groq, Mistral AI, Google Gemini), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Embeddings (Hugging Face Sentence-Transformers), Prompt Engineering
- **AI Agents & Automation:** Tool-calling AI agents (LangChain create_agent), autonomous tool selection & reasoning, external API integration (Tavily, OpenWeatherMap), agent-style automation patterns, n8n workflow automation, rules-based database triggers
- **Data Analytics & BI:** MS Excel (Advanced), Power BI, Data Cleaning, Data Visualization, KPI Dashboards
- **Database:** MySQL, SQLAlchemy, Database Triggers & Views
- **Tools & Platforms:** GitHub, VS Code, Streamlit Community Cloud

PROJECTS

CityPilot — Tool-Calling AI Agent for City Weather & News
Python · LangChain · Google Gemini · Tavily · OpenWeatherMap API · Streamlit *[GitHub-Link]*

- Built a tool-calling AI agent that autonomously decides whether to call a live-weather tool, a news-search tool, or both in a single turn — based entirely on reasoning over the user's question, with no hardcoded routing logic.
- Integrated OpenWeatherMap and Tavily as LangChain tools inside a create_agent tool-calling loop, using Google Gemini as the underlying reasoning model.
- Designed and built a custom Streamlit interface with a transparent "tool activity" panel that surfaces every tool call, its arguments, and its result — making the agent's reasoning auditable rather than a black box.

Marginalia — Chat with your PDF (RAG Chatbot)
Python · LangChain · ChromaDB · Hugging Face Embeddings · Mistral AI · Streamlit *[GitHub-Link]*

- Built and deployed a full-stack Retrieval-Augmented Generation (RAG) application that lets users upload PDFs and ask natural-language questions, with every answer grounded strictly in retrieved document content.
- Engineered the ingestion pipeline — PDF parsing, recursive chunking, and Hugging Face sentence-transformer embeddings — feeding a persistent ChromaDB vector store, with MMR-based retrieval tuned for relevance and diversity.
- Orchestrated the RAG chain with LangChain (ChatPromptTemplate, ChatMistralAI) and shipped a production-style Streamlit UI featuring session-isolated document indexing, secure API-key handling, and per-answer source citations.

ReelParse 2026
Python · LangChain · Mistral AI (mistral-small-2506) · Pydantic · Streamlit *[GitHub-Link]*

- Built an AI-powered movie information extractor that parses an unstructured paragraph of text and returns a clean, schema-validated structured record (title, year, genre, director, cast, rating, summary).
- Designed a Pydantic schema (Movie model) with PydanticOutputParser to enforce validated, typed output from the LLM — replacing a fragile string-matching parser with a robust, production-realistic pattern for consuming LLM output.
- Orchestrated prompting and response parsing with LangChain (ChatPromptTemplate, ChatMistralAI) and shipped both a Streamlit web UI and a CLI version for interactive and scripted use.

CERTIFICATIONS & TRAINING

Data Analytics with Generative AI **Apr 2026 – Jul 2026**

- Centre of Excellence (COE), Anudip Foundation, Kolkata — Trained in combining data analytics techniques with Generative AI tools to clean, analyze, and derive insights from datasets.

Ethical Hacking & Penetration Testing Intern **Aug 2024 – Oct 2024**

- C-DAC (Centre for Development of Advanced Computing), Noida — Applied cybersecurity concepts including ethical hacking methodologies and penetration testing on real-world systems; documented findings and recommended security improvements.

EDUCATION

B.Tech – Information Technology2022 – 2026
Government College of Engineering and Leather Technology, Kolkata | CGPA: 7.69/10.0
Class XII – Science (WBCHSE), 2020: 78.4% | Class X (WBBSE), 2018: 80.4%

AWARDS & LANGUAGES

Award: 1st Rank – English Debate Competition, Xbridge (2024) | Languages: English (Fluent), Bengali (Fluent), Hindi (Elementary)